

3. Homomorphism and isomorphism

3.1. HOMOMORPHISM AND ISOMORPHISM

READING. [Moore, §4, pp. 27–35].

Chapter 2 produced several groups that were visibly “the same”: reflection and particle exchange both gave $\mathbb{Z}_2$; $SO(2)$ and $U(1)$ had identical group laws; D_2 , V_4 and $\mathbb{Z}_2 \times \mathbb{Z}_2$ kept reappearing. This chapter makes “the same” precise.

homomorphism
同态

Definition 3.1 (Homomorphism). Let $(G, \mathbf{m}, \mathbf{I}, e)$ and $(G', \mathbf{m}', \mathbf{I}', e')$ be groups. A map $\varphi : G \rightarrow G'$ is a *homomorphism* if

$$\varphi(\mathbf{m}(g_1, g_2)) = \mathbf{m}'(\varphi(g_1), \varphi(g_2)), \quad \text{i.e.} \quad \varphi(g_1 g_2) = \varphi(g_1) \varphi(g_2)$$

for all $g_1, g_2 \in G$.

Equivalently, the square

$$\begin{array}{ccc} G \times G & \xrightarrow{\mathbf{m}} & G \\ \varphi \times \varphi \downarrow & & \downarrow \varphi \\ G' \times G' & \xrightarrow{\mathbf{m}'} & G' \end{array}$$

commutative diagram
交换图表

commutes.

The definition asks only that φ respect the multiplication. Nothing is required of the identity or of the inversion, and nothing need be, because both follow from it.

From $\varphi(e) = \varphi(e e) = \varphi(e) \varphi(e)$, multiplying by $\varphi(e)^{-1}$ gives $\varphi(e) = e'$. Then $e' = \varphi(e) = \varphi(g g^{-1}) = \varphi(g) \varphi(g^{-1})$, so $\varphi(g^{-1}) = [\varphi(g)]^{-1}$.

injective
单射

surjective
满射

Definition 3.2 (Injective, surjective, bijective). A map $\varphi : G \rightarrow G'$ is *injective* if $\varphi(g_1) = \varphi(g_2)$ forces $g_1 = g_2$, *surjective* if every $g' \in G'$ equals $\varphi(g)$ for some $g \in G$, and *bijective* if it is both.

bijective
双射

For a homomorphism, injectivity has to be tested at one point only.

Proposition 3.3. A homomorphism $\varphi : G \rightarrow G'$ is injective if and only if $\varphi(g) = e'$ forces $g = e$.

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Proof. If φ is injective, then $\varphi(g) = e' = \varphi(e)$ gives $g = e$. Conversely, suppose the condition holds and $\varphi(g_1) = \varphi(g_2)$. Then

$$\varphi(g_1 g_2^{-1}) = \varphi(g_1) \varphi(g_2)^{-1} = e',$$

so $g_1 g_2^{-1} = e$ and $g_1 = g_2$. □

isomorphism

同构

automorphism

自同构

Definition 3.4 (Isomorphism and automorphism). A bijective homomorphism is an *isomorphism*. If there is an isomorphism $G \rightarrow G'$ the two groups are *isomorphic*, written $G \cong G'$. An isomorphism from a group to itself is an *automorphism*, and the automorphisms of G are collected in $\text{Aut}(G)$.

The definition of $\cong$ runs one way, from G to G' . For “the same” to be a reasonable thing to say, it should not matter which way round.

Proposition 3.5. *If φ is an isomorphism then so is φ^{-1} , and $\cong$ is an equivalence relation.*

Proof. The inverse of a bijection is a bijection, so the only thing to check is that φ^{-1} respects multiplication. Take any $g'_1, g'_2 \in G'$. Since φ is surjective there are $g_1, g_2 \in G$ with $\varphi(g_i) = g'_i$, and then

$$\varphi^{-1}(g'_1 g'_2) = \varphi^{-1}(\varphi(g_1) \varphi(g_2)) = \varphi^{-1}(\varphi(g_1 g_2)) = g_1 g_2 = \varphi^{-1}(g'_1) \varphi^{-1}(g'_2).$$

For the second statement, the identity map on G is an isomorphism, so $G \cong G$. If φ gives $G \cong G'$, then φ^{-1} gives $G' \cong G$ by what has just been shown. And if φ gives $G \cong G'$ and ψ gives $G' \cong G''$, then $\psi\varphi$ is a bijection and $\psi\varphi(g_1 g_2) = \psi(\varphi(g_1) \varphi(g_2)) = \psi\varphi(g_1) \psi\varphi(g_2)$, so $G \cong G''$. □

Isomorphic groups are treated as the same group, which is what licenses $SO(2) \cong U(1)$ and $D_2 \cong V_4$ in Chapter 2.

Proposition 3.6. *$\text{Aut}(G)$ is a group under composition.*

Proof. The identity map is an automorphism. If $\varphi, \psi \in \text{Aut}(G)$ then $\psi\varphi$ is a bijection and respects multiplication, as in the proof of Proposition 3.5, so $\psi\varphi \in \text{Aut}(G)$, and $\varphi^{-1} \in \text{Aut}(G)$ by that proposition. Associativity is that of composition. □

Example 3.7 (Multiplication maps on $\mathbb{Z}_N$). Fix an integer k and define $m_k : \mathbb{Z}_N \rightarrow \mathbb{Z}_N$ by

$$m_k(\bar{x}) := \overline{kx}.$$

It is well defined, since replacing x by $x + N$ changes kx by kN , and it is a homomorphism, since $k(x + y) = kx + ky$. Taking $k = 1$ gives the identity map, $k = 0$ the map sending everything to $\bar{0}$, and $m_{k+N} = m_k$.

For which k is m_k an automorphism? A map of a finite set to itself is bijective exactly when it is injective, so by Proposition 3.3 the question is whether $N \mid kx$ forces $N \mid x$.

Suppose first that $d = \gcd(k, N) > 1$, and take $x = N/d$. Then $0 < x < N$, so $\bar{x} \neq \bar{0}$, while $kx = (k/d)N$ is a multiple of N , so $m_k(\bar{x}) = \bar{0}$. The map is not injective.

Suppose instead that $\gcd(k, N) = 1$, so no prime dividing N divides k . If $N \mid kx$, then every prime power in N must divide x , none of it being available from k , and so $N \mid x$. Hence m_k is an automorphism exactly when

$$\gcd(k, N) = 1.$$

For $N = 4$ the map m_3 sends $\bar{0}, \bar{1}, \bar{2}, \bar{3}$ to $\bar{0}, \bar{3}, \bar{2}, \bar{1}$, exchanging $\bar{1}$ and $\bar{3}$, whereas m_2 sends them to $\bar{0}, \bar{2}, \bar{0}, \bar{2}$ and collapses $\mathbb{Z}_4$ onto the subgroup $\{\bar{0}, \bar{2}\}$, which is a copy of $\mathbb{Z}_2$ since $\bar{2} + \bar{2} = \bar{0}$.

Nothing else occurs. An automorphism of $\mathbb{Z}_N$ is fixed by the image of $\bar{1}$, since $\bar{x} = x\bar{1}$ forces $\varphi(\bar{x}) = x\varphi(\bar{1})$, so every one of them is some m_k and

$$\text{Aut}(\mathbb{Z}_N) = \{m_k : \gcd(k, N) = 1\}.$$

KEY POINTS

- A homomorphism is a map that respects multiplication; respecting the identity and inverses then comes for free.
- A homomorphism is injective exactly when $\varphi(g) = e'$ forces $g = e$.
- An isomorphism is a bijective homomorphism, its inverse is again one, and isomorphic groups are treated as the same group.
- $\text{Aut}(G)$, the automorphisms of G , is a group under composition, and $\text{Aut}(\mathbb{Z}_N)$ consists of the maps $\bar{x} \mapsto \overline{kx}$ with $\gcd(k, N) = 1$.

3.2. KERNEL AND IMAGE

READING. [Moore, §4.1, pp. 35–37].

A homomorphism can fail to be injective, and it can fail to be surjective. Two subgroups measure the two failures.

Definition 3.8 (Kernel and image). Let $\varphi : G \rightarrow G'$ be a homomorphism. Its *kernel* and *image* are

$$\ker \varphi := \{g \in G : \varphi(g) = e'\} \subset G, \quad \text{im } \varphi := \varphi(G) \subset G'.$$

Proposition 3.9. $\ker \varphi$ is a subgroup of G , and $\text{im } \varphi$ is a subgroup of G' .

kernel
核
image
像

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Proof. Both contain the identity, by $\varphi(e) = e'$, so both are nonempty. If $g_1, g_2 \in \ker \varphi$ then

$$\varphi(g_1 g_2^{-1}) = \varphi(g_1) \varphi(g_2)^{-1} = e',$$

so $g_1 g_2^{-1} \in \ker \varphi$. If $g'_1 = \varphi(g_1)$ and $g'_2 = \varphi(g_2)$ lie in $\text{im } \varphi$ then $g'_1 (g'_2)^{-1} = \varphi(g_1 g_2^{-1}) \in \text{im } \varphi$. Apply Lemma 2.5 to each. $\square$

Proposition 3.10. φ is injective if and only if $\ker \varphi = \{e\}$. Hence φ is an isomorphism if and only if

$$\ker \varphi = \{e\} \quad \text{and} \quad \text{im } \varphi = G'.$$

Proof. The first statement is Proposition 3.3 written in the present notation, and the second combines it with the definition of $\text{im } \varphi$. $\square$

Quiz 3.1. Show that $\ker \varphi$ absorbs conjugation, that is $gkg^{-1} \in \ker \varphi$ for every $k \in \ker \varphi$ and every $g \in G$. Subgroups with this property are the subject of Chapter 6.

3.3. EXAMPLES

Example 3.11 ($\mathbb{Z}_N \cong \mu_N$). Define $\varphi : \mathbb{Z}_N \rightarrow \mu_N$ by $\varphi(\bar{r}) = e^{2\pi i r/N}$. Replacing r by $r + N$ does not change the value, so φ is well defined on residue classes, and

$$\varphi(\bar{r}_1 + \bar{r}_2) = e^{2\pi i(r_1 + r_2)/N} = \varphi(\bar{r}_1) \varphi(\bar{r}_2)$$

makes it a homomorphism from addition to multiplication. Its kernel is $\{\bar{0}\}$ and its image is all of μ_N , so $\mathbb{Z}_N \cong \mu_N$ by Proposition 3.10. The residue classes of §2.3 and the roots of unity are the same group written twice.

Example 3.12 (The power map). For $k \in \mathbb{Z}$ let $P_k : \mu_N \rightarrow \mu_N$ be $z \mapsto z^k$. It is a homomorphism, since $(z_1 z_2)^k = z_1^k z_2^k$. Its kernel is

$$\ker P_k = \{z \in \mu_N : z^k = 1\} = \mu_d, \quad d = \gcd(k, N),$$

so P_k is an isomorphism exactly when $\gcd(k, N) = 1$, which is Example 3.7 again in multiplicative notation. At the other extreme P_N is the trivial map. For $N = 4$ and $k = 2$,

$$\mu_4 = \{1, i, -1, -i\}, \quad \ker P_2 = \{\pm 1\} \cong \mathbb{Z}_2, \quad \text{im } P_2 = \{\pm 1\} \cong \mu_2.$$

Example 3.13 ($U(1) \rightarrow SU(2)$). Fix an integer $N \geq 2$ and send $z \in U(1)$ to

$$\varphi(z) = \begin{pmatrix} z^N & 0 \\ 0 & z^{-N} \end{pmatrix}.$$

The matrix is unitary with determinant 1, so φ maps into $SU(2)$, and $\varphi(z_1 z_2) = \varphi(z_1)\varphi(z_2)$. Here

$$\ker \varphi = \{z : z^N = 1\} = \mu_N, \quad \text{im } \varphi = \left\{ \begin{pmatrix} w & 0 \\ 0 & w^{-1} \end{pmatrix} : |w| = 1 \right\} \cong U(1),$$

so this φ is neither injective nor surjective, and the two subgroups say by how much it misses.

Example 3.14 ($SO(2) \cong U(1)$). Example 2.19 recorded the correspondence $R(\phi) \leftrightarrow e^{i\phi}$ and called the two groups the same. The map behind that statement is

$$\varphi : SO(2) \rightarrow U(1), \quad \varphi(R(\phi)) = e^{i\phi}.$$

It is well defined, since $R(\phi)$ fixes ϕ only up to a multiple of 2π , which leaves $e^{i\phi}$ unchanged, and it is a homomorphism, since $R(\phi_1)R(\phi_2) = R(\phi_1 + \phi_2)$. Its kernel is $\{1\}$, because $e^{i\phi} = 1$ forces $\phi \in 2\pi\mathbb{Z}$ and $R(2\pi n) = 1$, and its image is all of $U(1)$. So $SO(2) \cong U(1)$ by Proposition 3.10: a rotation of the plane and a phase are the same datum.

3.4. $SU(2)$ AND $SO(3)$

READING. [Moore, §5.4.1, pp. 45–48].

Recall from §2.5 that $SO(3)$ is the group of proper rotations of $\mathbb{R}^3$. We now build a homomorphism $SU(2) \rightarrow SO(3)$ and find it two-to-one, with u and $-u$ giving the same rotation of space.

To make $SU(2)$ act on $\mathbb{R}^3$ we look for a three-dimensional real space of matrices preserved by conjugation, conjugation being what a group of matrices does naturally to matrices. A trace survives any conjugation, and conjugation by a unitary preserves Hermiticity, which makes traceless and Hermitian the condition to try. The 2×2 matrices meeting it are exactly the real combinations $x_i \sigma^i$ of the Pauli matrices of §2.6. There are three of those, so the space is three-dimensional over $\mathbb{R}$, and it is familiar ground: the spin operators $S^i = \frac{\hbar}{2} \sigma^i$ live there, and for a unit vector $\hat{n}$ the combination $\frac{\hbar}{2} \hat{n} \cdot \vec{\sigma}$ is the spin component along $\hat{n}$.

Write $\mathcal{H}$ for that space and send a vector into it by

$$h : \mathbb{R}^3 \rightarrow \mathcal{H}, \quad h(\vec{x}) = \vec{x} \cdot \vec{\sigma} = x_i \sigma^i = \begin{pmatrix} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & -x_3 \end{pmatrix}.$$

This is a linear bijection, and it carries lengths along with it, stored in the determinant as

$$\det h(\vec{x}) = -x_3^2 - (x_1^2 + x_2^2) = -|\vec{x}|^2.$$

which is how the construction will see that rotations preserve them.

For $u \in SU(2)$ let $C_u(m) = umu^\dagger$. This is the conjugation promised above, and it does map $\mathcal{H}$ to itself, since $\text{Tr}(umu^\dagger) = \text{Tr}(m) = 0$ and $(umu^\dagger)^\dagger = umu^\dagger$.

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Definition 3.15 (The rotation attached to u). $R(u) := h^{-1} \circ C_u \circ h$, that is, $R(u)$ is the map $\mathbb{R}^3 \rightarrow \mathbb{R}^3$ determined by

$$u(\vec{x} \cdot \vec{\sigma})u^\dagger = (R(u)\vec{x}) \cdot \vec{\sigma} \quad \text{for all } \vec{x} \in \mathbb{R}^3, \quad \text{equivalently} \quad u\sigma^i u^\dagger = R(u)_{ji}\sigma^j.$$

In a composite the rightmost map acts first, so $R(u)$ carries $\vec{x}$ into $\mathcal{H}$ as $\vec{x} \cdot \vec{\sigma}$, conjugates that matrix by u , and brings the result back by h^{-1} . It begins and ends in $\mathbb{R}^3$, and all three steps are linear bijections, so $R(u)$ is a linear bijection of $\mathbb{R}^3$.

The displayed equation says this without naming h . Conjugation keeps $u(\vec{x} \cdot \vec{\sigma})u^\dagger$ inside $\mathcal{H}$, so some vector of $\mathbb{R}^3$ has it as image, and h is injective, so only one does. That vector is $R(u)\vec{x}$, and it is the uniqueness that makes the equation a definition rather than a condition. Putting $\vec{x} = \hat{e}_i$ gives the index form.

Both readings live in the square

$$\begin{array}{ccc} \mathbb{R}^3 & \xrightarrow{R(u)} & \mathbb{R}^3 \\ h \downarrow \sim & & \sim \downarrow h \\ \mathcal{H} & \xrightarrow{C_u} & \mathcal{H} \end{array}$$

which commutes, the $\sim$ marking the two bijections. Definition 3.15 runs down, along, and back up, its last leg being h^{-1} . The equation instead compares the two routes from the top left to the bottom right, down then along against along then down, and never inverts h .

Theorem 3.16. $R : SU(2) \rightarrow SO(3)$ is a surjective homomorphism with $\ker R = \{\pm 1\} \cong \mathbb{Z}_2$.

Proof. Homomorphism. Conjugating twice,

$$(R(u_1 u_2)\vec{x}) \cdot \vec{\sigma} = (u_1 u_2)(\vec{x} \cdot \vec{\sigma})(u_1 u_2)^\dagger = u_1 [(R(u_2)\vec{x}) \cdot \vec{\sigma}] u_1^\dagger = (R(u_1)R(u_2)\vec{x}) \cdot \vec{\sigma},$$

and h is injective, so $R(u_1 u_2) = R(u_1)R(u_2)$.

Orthogonal. Conjugation does not change a determinant, so

$$|R(u)\vec{x}|^2 = -\det((R(u)\vec{x}) \cdot \vec{\sigma}) = -\det(u(\vec{x} \cdot \vec{\sigma})u^\dagger) = -\det(\vec{x} \cdot \vec{\sigma}) = |\vec{x}|^2,$$

and $R(u) \in O(3)$.

Determinant. Write the product rule with a summed index ℓ , as $\sigma^i \sigma^j = \delta^{ij} \mathbb{1} + i\epsilon^{ij\ell} \sigma^\ell$, multiply on the right by σ^k , and take the trace. With $\text{Tr } \sigma^k = 0$ and $\text{Tr}(\sigma^\ell \sigma^k) = 2\delta^{\ell k}$,

$$\text{Tr}(\sigma^i \sigma^j \sigma^k) = 2i\epsilon^{ijk}.$$

The factors $u^\dagger u$ inserted between the three matrices cancel under the trace, so $\text{Tr}(\sigma^1 \sigma^2 \sigma^3) = \text{Tr}((u\sigma^1 u^\dagger)(u\sigma^2 u^\dagger)(u\sigma^3 u^\dagger))$. Substituting $u\sigma^a u^\dagger = R(u)_{ia}\sigma^i$,

$$\begin{aligned} 2i &= \text{Tr}(\sigma^1 \sigma^2 \sigma^3) = R(u)_{i1} R(u)_{j2} R(u)_{k3} \text{Tr}(\sigma^i \sigma^j \sigma^k) \\ &= 2i\epsilon^{ijk} R(u)_{i1} R(u)_{j2} R(u)_{k3} = 2i \det R(u). \end{aligned}$$

So $\det R(u) = 1$ and $R(u) \in SO(3)$.

Kernel. $R(u) = 1$ says $u\sigma^i u^\dagger = \sigma^i$ for each i , so u commutes with $\sigma^1, \sigma^2, \sigma^3$ and therefore with every 2×2 matrix, those three together with 1 being a basis. By Proposition 2.17, $u = \lambda 1$, and $\det u = \lambda^2 = 1$ forces $\lambda = \pm 1$. Both signs do lie in the kernel, since $(\pm 1)m(\pm 1)^\dagger = m$ for every m , so

$$\ker R = \{1, -1\} \cong \mathbb{Z}_2.$$

Surjective. Take $u = \exp(-\frac{i}{2}\theta\sigma^3) = \text{diag}(e^{-i\theta/2}, e^{i\theta/2})$. Then

$$u\sigma^1 u^\dagger = \begin{pmatrix} 0 & e^{-i\theta} \\ e^{i\theta} & 0 \end{pmatrix} = \cos \theta \sigma^1 + \sin \theta \sigma^2, \quad u\sigma^3 u^\dagger = \sigma^3,$$

so $R(u)$ is the rotation by θ about the third axis. Starting from σ^2 instead gives the rotation by θ about the second axis. Write $R_3(\theta)$ and $R_2(\theta)$ for these, both of them in the image of R .

Every rotation is a product of three of them. Take $A \in SO(3)$ and let β and α be the polar and azimuthal angles of $A\hat{e}_3$, so that $Q := R_3(\alpha)R_2(\beta)$ sends $\hat{e}_3$ to $A\hat{e}_3$. Then $Q^{-1}A$ leaves $\hat{e}_3$ fixed and, being orthogonal, maps the perpendicular plane to itself. On that plane it has determinant 1, so it is a rotation of it, say $R_3(\gamma)$. Hence

$$A = R_3(\alpha)R_2(\beta)R_3(\gamma),$$

the *Euler angles* of A .

Euler angles
欧拉角

Since R is a homomorphism, a matrix producing A is the matching product,

$$u(\alpha, \beta, \gamma) := e^{-i\alpha\sigma^3/2} e^{-i\beta\sigma^2/2} e^{-i\gamma\sigma^3/2} = \begin{pmatrix} e^{-i(\alpha+\gamma)/2} \cos \frac{\beta}{2} & -e^{-i(\alpha-\gamma)/2} \sin \frac{\beta}{2} \\ e^{i(\alpha-\gamma)/2} \sin \frac{\beta}{2} & e^{i(\alpha+\gamma)/2} \cos \frac{\beta}{2} \end{pmatrix}, \quad (3.1)$$

which has $R(u(\alpha, \beta, \gamma)) = A$. So R is surjective, and letting $\alpha \in [0, 2\pi)$, $\beta \in [0, \pi]$ and $\gamma \in [0, 4\pi)$ reaches every element of $SU(2)$ this way. $\square$

Remark 3.17 (Double cover). Since $\ker R = \{\pm 1\}$, each rotation has exactly two lifts, u and $-u$. The two minus signs cancel in $u m u^\dagger$, which is why both give the same rotation.

The exponential $\exp(-\frac{i}{2}\theta\sigma^3)$ of the proof is the case $\hat{n} = \hat{e}_3$ of a formula good for any axis. Since $(\hat{n} \cdot \vec{\sigma})^2 = 1$ for a unit vector $\hat{n}$, expanding the exponential gives

$$u(\hat{n}, \theta) = \exp\left(-\frac{i}{2}\theta \hat{n} \cdot \vec{\sigma}\right) = \cos \frac{\theta}{2} 1 - i \sin \frac{\theta}{2} \hat{n} \cdot \vec{\sigma},$$

so $u(\hat{n}, 2\pi) = -1$ and $u(\hat{n}, 4\pi) = 1$. A rotation by 2π returns every vector to where it started, and only at 4π does u itself return.

The half angle is what produces the sign, and it is present in both descriptions of u . The axis-angle form above carries it as $\theta/2$. The Euler form (3.1) has a half angle in each of its three factors, and the domain fixed in the proof lets γ carry the sign: A is unchanged when γ advances by 2π , while $u(\alpha, \beta, \gamma + 2\pi) = -u(\alpha, \beta, \gamma)$, so for fixed α and β the period in γ is 4π for u and 2π for A .

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Remark 3.18 (The sign cannot be chosen away). Since R is two-to-one, one might hope to pick one lift for each rotation and keep a homomorphism, that is, to find $s : SO(3) \rightarrow SU(2)$ with $R \circ s$ the identity. There is none. Let A be the rotation by π about the third axis, so that $A^2 = \mathbb{1}$. Its two lifts are $\pm u$ with $u = u(\hat{e}_3, \pi) = -i\sigma^3$, and both square to $(\pm u)^2 = -\mathbb{1}$. A homomorphism would need $s(A)^2 = s(A^2) = \mathbb{1}$ instead, so no such s exists.

Remark 3.19 (The belt). The factor of two is visible in $SO(3)$ itself. Hold one end of a belt fixed and turn the other end through 2π . The twist cannot be worked out while both ends are held. Turn it through 4π instead and it can. The 2π rotation cannot be deformed continuously into no rotation with the ends held fixed, while the 4π one can. In the jargon, the closed paths in $SO(3)$ that start and end at the identity fall into two classes under such deformation, that is, the fundamental group of $SO(3)$ is

$$\pi_1(SO(3)) \cong \mathbb{Z}_2,$$

and an element of $SU(2)$ records a rotation together with the deformation class of a path reaching it from the identity. The demonstration is due to Dirac [Newman]; see [Staley] for an account written for physicists.

The same two-ness shows up in the shapes. $SU(2) \sim S^3$ by Example 2.20, and R identifies the antipodal pair $\pm u$, so $SO(3)$ is the three-sphere with antipodes identified, the real projective space $\mathbb{RP}^3$. Students who want the topology done properly will find it in [Nakahara, §4].

Group theory does not say which of the two a physical object uses. In quantum mechanics a spin- $\frac{1}{2}$ wavefunction is a two-component complex column, and u and $-u$ act on it with opposite signs although they induce the same rotation of space.

3.5. $GL(V)$ AND $GL(n, \kappa)$

Example 3.20 (Coordinates as an isomorphism). Let V be an n -dimensional vector space over κ and let $GL(V)$ be the group of invertible linear maps $V \rightarrow V$. An ordered basis $b = \{\hat{e}_1, \dots, \hat{e}_n\}$ turns each $\tau \in GL(V)$ into a matrix $T_b(\tau)$ through

$$\tau(\hat{e}_i) = \sum_j \hat{e}_j T_b(\tau)_{ji}.$$

Applying τ_1 after τ_2 to $\vec{v} = \sum_i v_i \hat{e}_i$,

$$\tau_1(\tau_2 \vec{v}) = \sum_{i,j} v_i T_b(\tau_2)_{ji} \tau_1(\hat{e}_j) = \sum_{i,j,k} \hat{e}_k T_b(\tau_1)_{kj} T_b(\tau_2)_{ji} v_i.$$

But applying τ_1 after τ_2 is applying $\tau_1 \tau_2$, and the defining formula used on $\tau_1 \tau_2$ itself reads

$$(\tau_1 \tau_2)(\vec{v}) = \sum_{i,k} \hat{e}_k T_b(\tau_1 \tau_2)_{ki} v_i.$$

The two right-hand sides agree for every $\vec{v}$, and the $\hat{e}_k$ are independent, so

$$T_b(\tau_1 \tau_2) = T_b(\tau_1) T_b(\tau_2)$$

and $\varphi_b : \tau \mapsto T_b(\tau)$ is a homomorphism $GL(V) \rightarrow GL(n, \kappa)$. It is surjective, since every invertible matrix defines an invertible map through the same formula. It is injective, since $T_b(\tau) = 1_n$ says $\tau(\hat{e}_i) = \hat{e}_i$ for every i and hence $\tau = \text{id}$. So

$$GL(V) \cong GL(n, \kappa).$$

The isomorphism depends on the basis. A second basis $\hat{e}'_i = \sum_j S_{ji} \hat{e}_j$ gives $T_{b'}(\tau) = S T_b(\tau) S^{-1}$, so the two coordinate pictures differ by conjugation. This is why an abstract group can be studied through matrices at all, and it is where representation theory starts.

A homomorphism into $GL(n, \kappa)$ is a way of realising an abstract group by matrices, and the same group can be realised in several ways.

Example 3.21 (Matrix realisations). The group $\mathbb{Z}$. For $a \in \kappa$ the map

$$\varphi_a : \mathbb{Z} \rightarrow GL(2, \kappa), \quad n \mapsto \begin{pmatrix} 1 & 0 \\ na & 1 \end{pmatrix},$$

is a homomorphism, since $\begin{pmatrix} 1 & 0 \\ na & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ ma & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ (n+m)a & 1 \end{pmatrix}$. For $a \neq 0$ it is injective, so $\mathbb{Z}$ is isomorphic to $\text{im } \varphi_a$, the group of all such matrices. For $a = 0$ every n goes to 1_2 and $\ker \varphi_0 = \mathbb{Z}$.

The group $S_2 = \{e, \sigma\}$ with $\sigma^2 = e$. The map $\varphi : S_2 \rightarrow GL(2, \kappa)$ with

$$\varphi(e) = 1_2, \quad \varphi(\sigma) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

is a homomorphism, since $\varphi(\sigma)^2 = 1_2$. It is injective, so S_2 is isomorphic to $\text{im } \varphi = \{1_2, \varphi(\sigma)\}$, a subgroup of order two inside $GL(2, \kappa)$. Since $S_2 \cong \mu_2 \cong \mathbb{Z}_2$, the same two matrices realise those groups as well.

The group $\mu_3 = \langle \omega \mid \omega^3 = 1 \rangle$. Sending ω to the cyclic permutation matrix

$$\varphi(\omega) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad \varphi(\omega^2) = \varphi(\omega)^2 = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad \varphi(1) = 1_3,$$

defines a homomorphism $\varphi : \mu_3 \rightarrow GL(3, \kappa)$, since $\varphi(\omega)^3 = 1_3$, and the three matrices are distinct, so φ is injective. The same group is realised in one dimension by $\psi : \mu_3 \rightarrow GL(1, \mathbb{C})$, $\omega^k \mapsto \omega^k$, again injective. One group, two matrix realisations, of different sizes.

The group D_4 . With $D_4 = \langle A, B \mid A^4 = B^2 = (AB)^2 = 1 \rangle$ of Example 2.24, put

$$\varphi(A) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \varphi(B) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

3. Homomorphism and isomorphism

The images satisfy the relations, $\varphi(A)^4 = \varphi(B)^2 = (\varphi(A)\varphi(B))^2 = 1_2$, so the assignment extends to a homomorphism $\varphi : D_4 \rightarrow GL(2, \mathbb{R})$. Its image

$$\text{im } \varphi = \{\pm 1_2, \pm\varphi(A), \pm\varphi(B), \pm\varphi(A)\varphi(B)\}$$

has eight elements, so φ is injective and D_4 is isomorphic to this group of matrices, the symmetries of the square written in coordinates. Sending both A and B to 1_2 is also a homomorphism $D_4 \rightarrow GL(2, \mathbb{R})$, with kernel all of D_4 .

Chapter 7 takes up the general question of which matrix realisations a group admits.

KEY POINTS

- $\ker \varphi$ and $\text{im } \varphi$ are subgroups, and they measure the failure of φ to be injective and surjective. φ is an isomorphism exactly when $\ker \varphi = \{e\}$ and $\text{im } \varphi = G'$.
- $R : SU(2) \rightarrow SO(3)$ is onto with kernel $\{\pm 1\}$, so every rotation has exactly two preimages $\pm u$ and $SU(2)$ is a double cover of $SO(3)$.
- $SU(2)$ is the three-sphere S^3 , and identifying the pairs $\pm u$ leaves $SO(3) \sim \mathbb{R}P^3$.
- Choosing a basis gives $GL(V) \cong GL(n, \kappa)$, and changing the basis conjugates the matrices.
- A homomorphism $G \rightarrow GL(n, \kappa)$ realises G by matrices, and an injective one makes G isomorphic to a group of matrices.

KEY TERMS. homomorphism 同态 · isomorphism 同构 · automorphism 自同构 · injective 单射 · surjective 满射 · bijective 双射 · kernel 核 · image 像 · double cover 双覆盖

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